

Six-Functor Formalisms over Orbit Categories

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Chapter 1

Introduction

Our Main idea come from [\[Sch26b\]](#) and [\[Sch26a\]](#).

Chapter 2

Stable Homotopy Theory

Chapter 3

Equivariant Homotopy Theory and Orbit Category

Let G denote a compact Lie group.

Definition 3.0.1. The orbit category $\mathcal{O}_G$ associated to G is defined by

Objects $\{G/H \mid H \text{ is a closed subgroup of } G\}$

Morphisms Continuous G -equivariant maps.

Let $\text{Hom}_{\mathcal{O}_G}(G/H, G/K)$ equipped with compact-open topology, then we have the homeomorphism

$$\text{Hom}_{\mathcal{O}_G}(G/H, G/K) \simeq (G/K)^H.$$

Definition 3.0.2. An $\mathcal{O}_G$ -space is a contravariant functor from $\mathcal{O}_G$ to Top .

Definition 3.0.3. A G -space is a covariant functor from classifying category BG to Top .

Theorem 3.0.4 (Elmendorf's Theorem).

Chapter 4

Six-Functor Formalisms

Throughout this chapter, we let $\mathcal{C}$ denote a category satisfying the following hypotheses

- The category $\mathcal{C}$ has all finite limits
- The class of morphisms E contains all isomorphisms. E is stable under pullback, composition and diagonals.

A six-functor formalism is a functor $\mathcal{D} : \mathcal{C}^{\text{op}} \rightarrow \text{CAlg}(\text{Cat}_{\infty})$ sending any morphism f in $\mathcal{C}$ to a symmetric monoidal functor f^* in $\text{CAlg}(\text{Cat}_{\infty})$ and an assignment

$$f : X \rightarrow Y \quad \mapsto \quad f_! : \mathcal{D}(X) \rightarrow \mathcal{D}(Y)$$

for any $f \in E$ to a functor of symmetric monoidal ∞ -categories.¹ We call functor f^* the star pullback and $f_!$ shriek pushforward.

Additionally, we require that

- There exists functor f_* right adjoint to f^* , we call f_* star pushforward;
- There exists functor $f^!$ right adjoint to $f_!$, we call $f^!$ shriek pullback;
- For any $X \in \mathcal{C}$ and $A \in \mathcal{D}(X)$, the symmetric monoidal ∞ -category $\mathcal{D}(X)$ admits a adjoint pair of functor

$$- \otimes A \dashv \underline{\text{Hom}}(A, -),$$

we call it tensor product and internal Hom;

- The star pullback functor $f^* : \mathcal{D}(Y) \rightarrow \mathcal{D}(X)$ compatible with symmetric monoidal structure on $\mathcal{D}(X)$ and $\mathcal{D}(Y)$ and compatible with composition;
- The shriek pushforward functor $f_! : \mathcal{D}(X) \rightarrow \mathcal{D}(Y)$ compatible with composition and satisfying the base change and projection formula isomorphism.

Here the base change formula means if there exists a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & \lrcorner & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

over $\mathcal{C}$, then the diagram

$$\begin{array}{ccc} \mathcal{D}(X') & \xleftarrow{g'^*} & \mathcal{D}(X) \\ f'_! \downarrow & & \downarrow f_! \\ \mathcal{D}(Y') & \xleftarrow{g^*} & \mathcal{D}(Y) \end{array}$$

commutes up to homotopic coherent or equivariantly we say $f'_! \circ g'^*$ is natural isomorphic to $g^* \circ f_!$.

The projection formula means $\mathcal{D}(X)$ is equipped with $\mathcal{D}(Y)$ -module structure via $f^* : \mathcal{D}(Y) \rightarrow \mathcal{D}(X)$ and functor $f_!$ is $\mathcal{D}(Y)$ -linear. Equivalently, for any $A \in X$ and $B \in Y$ the shriek pushforward satisfies

$$f_!(A \otimes f^* B) = f_!(A) \otimes B.$$

¹ $f_!$ need not be a symmetric monoidal functor.

Example 4.0.1 (Six functor formalism for abelian sheaves). *Let S be the base scheme and τ denote a Grothendieck topology of Sch/S .² Define the functor $\mathcal{D} : (\mathrm{Sch}/S)^{\mathrm{op}} \rightarrow \mathrm{CAlg}(\mathrm{Cat}_{\infty})$ by assigning*

$$X \mapsto \mathcal{D}(\mathrm{Ab}(X_{\tau})) \quad (f : X \rightarrow Y) \mapsto (f^{-1} : \mathcal{D}(\mathrm{Ab}(Y_{\tau})) \mapsto \mathcal{D}(\mathrm{Ab}(X_{\tau})))$$

where $X \in \mathrm{Sch}/S$ and $\mathcal{D}(\mathrm{Ab}(X_{\tau}))$ denotes the derived ∞ -category of the abelian sheaves on X . Let E be the class of compactifiable morphisms, i.e., for any $f : X \rightarrow Y$ in E , there exists a factorization

$$f : X \xrightarrow{j} \bar{X} \xrightarrow{p} Y$$

such that j is an open immersion and p is a proper morphism. Define $f_! = Rp_ \circ j_!$ where $j_!$ is extension by zero.*

Example 4.0.2 (Six functor formalism for quasi-coherent sheaves). *Let S be the base scheme. Define the functor $\mathcal{D} : (\mathrm{Sch}/S)^{\mathrm{op}} \rightarrow \mathrm{CAlg}(\mathrm{Cat}_{\infty})$ by assigning*

$$X \mapsto \mathcal{D}_{\mathrm{qc}}(X) \quad f \mapsto Lf^*.$$

where $\mathcal{D}_{\mathrm{qc}}(X)$ denotes the derived ∞ -category of complexes with quasi-coherent cohomology sheaves on X .³ Let

$$E = \{\text{proper morphism in } (\mathrm{Sch}/S)_{\mathrm{qcqs}}\}$$

Define the shriek pushforward by $f_! := Rf_$. The existence of the right adjoint of $f_!$ is the statement of Grothendieck duality.*

²We can choose τ be the Zariski, Étale, Nisnevich, fpqc, or fppf topology.

³When X is qcqs, we have $\mathcal{D}_{\mathrm{qc}}(X) = \mathcal{D}(\mathrm{QCoh}(X))$.

Chapter 5

Gestalt

Bibliography

[Sch26a] Peter Scholze. Geometry and higher category theory, 2026. Lecture notes for course WS 2025/26.

[Sch26b] Peter Scholze. Six-functor formalisms, 2026.